

Decay Logic

Purpose of Decay

Reputation equals current trustworthiness.

If a user becomes inactive for a long period, the system should reduce confidence in that user just as it happens in real life interactions.

Time based decay ensures that reputation reflects recent behavior instead of **historical total**.

Time in Blockchain Systems

In Eth time is measured using `block.timestamp`.(remix)

Which is the Unix timestamp(sec) of the block in which a transaction is included.

- It is provided by the validator.
- It is approximately correct but not perfectly precise.
- It cannot be used for exact scheduling.
- It is sufficient for coarse time measurements such as days or weeks.

The system relies on this timestamp to calculate inactivity duration.

Why Continuous Decay Is Not Possible?

Smart contracts do not run automatically. A smart contract only executes when =>

- A user sends a transaction.
- A function is called.

Because of this =>

- The contract cannot reduce reputation every day automatically.
- Decay must be calculated only when a user interacts again.

This is a fundamental limitation of blockchain execution. Hence our first threat.

Discrete Decay Model

Decay is applied in fixed intervals. For example:

- Decay Interval: 30 days
- Decay Amount: 5 reputation points per interval

When an user interacts =>

1. Calculate time elapsed since last update.
2. Determine how many full intervals have passed.
3. Apply decay for each full interval.
4. Ensure reputation does not go below the minimum limit ofcourse!

initial formula =>

$$\text{Elapsed Time} / \text{Decay Interval} = \text{no. Decay Periods}$$

Only complete intervals count. Partial intervals are ignored. (**for now**)

OrderedOperations

When a user interacts →

1. Retrieve stored reputation.
2. Calculate time since last update.
3. Apply decay.
4. Apply interaction reward.
5. Update the stored timestamp.

Boundary Conditions

The system enforces the following limits=>

- Reputation cannot fall below 0.
- Reputation cannot exceed 100.
- If decay exceeds current reputation, reputation is set to 0.

Design Rationale

The decay logic is deterministic, simple to verify, transparent, suitable for smart contract execution. It does not attempt to model complex trust dynamics. It prioritizes clarity and correctness over mathematical showoff.